Module 8: Cellular Genetics — Study Questions

1. Name the three stages of interphase (G_1, S, G_2). What major event occurs during each?
2. After S phase, are the two copies of a chromosome called homologs or sister chromatids? How can you tell the difference?
3. List the four phases of mitosis in order (PMAT). Briefly describe what happens during each.
4. What structures separate during anaphase of mitosis?
5. At the end of mitosis and cytokinesis, how many daughter cells are produced? Are they diploid or haploid? Are they genetically identical to the parent cell?
6. Compare cytokinesis in animal cells (cleavage furrow) vs. plant cells (cell plate).
7. Where does meiosis occur in humans? What is the purpose of meiosis?
8. How many rounds of division occur in meiosis? How many daughter cells are produced? What is their ploidy?
9. What structures separate during Anaphase I? How does this differ from Anaphase II?
10. Describe synapsis and crossing over. During which phase do they occur? Why are they important?
11. What is independent assortment? How does random alignment of homologs at Metaphase I contribute to genetic diversity?
12. In humans ($2n = 46$), how many unique gamete combinations are possible from independent assortment alone?
13. Complete a comparison table: Mitosis vs. Meiosis (number of divisions, daughter cells, ploidy, genetic identity, purpose).

14. Why is it critical that gametes are haploid? What would happen if gametes were diploid?

15. Name the three sources of genetic variation in sexual reproduction. Which two occur during meiosis?

16. What is nondisjunction? During which stage(s) of meiosis can it occur?

17. What is aneuploidy? Give one example of an aneuploid condition in humans.

18. What are cell cycle checkpoints (G_1, G_2, M)? What is evaluated at each?

19. Distinguish between a proto-oncogene and an oncogene. What role does p53 play?

20. Explain why siblings from the same parents are not genetically identical to each other.